

HCA-Net: Hierarchical Contextual Attention Network for Lightweight and Accurate Polyp Segmentation

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Abstract—Early detection of colorectal polyps is crucial for clinical screening and cancer prevention, where accurate and efficient automatic segmentation plays a pivotal role. However, colonoscopy images often suffer from low contrast, blurred boundaries, and scale variations, making segmentation challenging. Existing encoder-decoder networks (e.g., U-Net) suffer from asymmetric supervision and feature redundancy, which in turn lead to semantic inconsistency and loss of fine details. While deeper or hybrid designs alleviate these issues, their high complexity and computational burden limit feasibility in real-time clinical practice. To address these challenges, we propose a lightweight segmentation framework, Hierarchical Contextual Attention Network (HCA-Net), consisting of the Redundancy-Suppressed Dual-Path Downsampling (RS-DPD) module and the Boundary-Aware Semantic Alignment Upsampling (BA-SAU) module, applied to the encoder and decoder, respectively. RS-DPD suppresses redundancy while preserving fine-grained details through a dual-path design, whereas BA-SAU leverages cross-layer contextual attention to enforce semantic consistency and enhance boundary sensitivity. Both modules are built upon our proposed Hierarchical Contextual Attention (HCA) mechanism,

which combines convolutional projection with pooling-based compression to achieve efficient global modeling and accurate local boundary restoration. In addition, a composite boundary-aware loss function is designed to improve pixel-level accuracy, structural consistency, and robustness in low-contrast and boundary-ambiguous regions. Extensive experiments on public colorectal polyp datasets demonstrate that HCA-Net achieves state-of-the-art (SOTA) segmentation accuracy with significantly improved efficiency, while maintaining robustness under low-contrast and blurred-boundary conditions.

Index Terms—Polyp segmentation, hierarchical contextual attention, dual-path downsampling, semantic alignment upsampling.

I. INTRODUCTION

C OLORECTAL cancer is one of the most common malignant tumors worldwide, and early screening is of great clinical significance for reducing both incidence and mortality rates [1]. Colonoscopy, as the primary screening method, enables direct observation of colorectal polyps; however, its effectiveness is highly dependent on the physician's experience. Polyps with low contrast, flat morphology, or blurred boundaries are prone to being missed or misdiagnosed [2]. Therefore, developing efficient and robust automatic segmentation algorithms is essential for advancing computer-aided diagnosis systems into clinical practice.

In recent years, deep learning has achieved remarkable progress in medical image segmentation. Encoder-decoder architectures, represented by U-Net [3], have become mainstream solutions owing to their hierarchical feature extraction and skip-connection mechanisms. These models progressively capture multi-scale semantic information during the encoding stage and gradually recover spatial resolution through feature fusion during the decoding stage, thereby yielding relatively high-quality segmentation with sharp boundaries. However, as illustrated in Fig. 1, feature visualization across different stages reveals two inherent limitations:

1) *Asymmetric supervision between encoder and decoder:*

As illustrated in Fig. 1(b), training mainly applies supervision at the decoder (final output or a few intermediate layers), leaving the encoder without direct semantic

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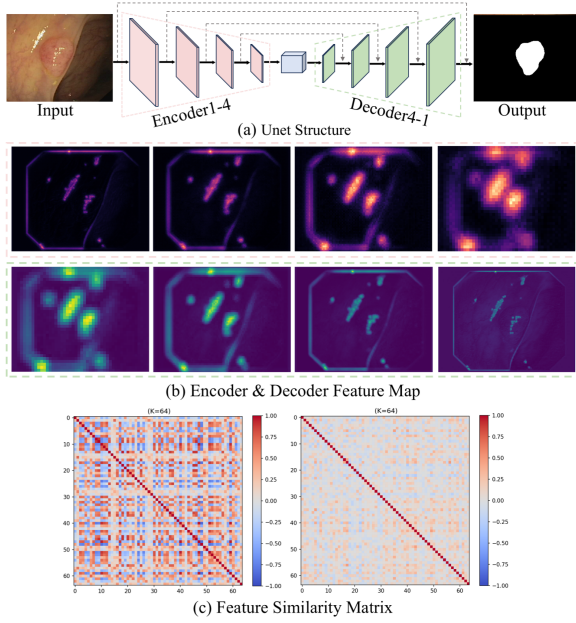


Fig. 1. Information Attenuation and Feature Redundancy in the U-Net Encoder-Decoder Architecture.

guidance. This weakens deep feature learning and attenuates semantic information during propagation, often causing boundary errors or missed detection of small polyps in low-contrast areas.

- 2) *Feature redundancy and insufficient discriminability*: As illustrated in Fig. 1(c), shallow layers capture rich edge and texture details, but repeated downsampling and convolutions in deeper layers introduce channel redundancy and reduce feature discriminability, causing fine-grained information to degrade. Skip connections partially recover lost details, but they are often insufficient for low-contrast or morphologically diverse polyps, leading to unstable predictions.

In traditional encoder-decoder frameworks, convolutional neural networks (CNNs) [4] are typically employed for multi-level feature extraction and boundary refinement, thereby improving representation capability and segmentation accuracy. However, their limited receptive fields restrict effective global context modeling [5], which results in insufficient semantic consistency in complex structures. To address this, Transformers have been introduced to capture long-range dependencies [6]. While they improve global context encoding, their high computational cost and weak ability in preserving fine details limit their adoption in medical image segmentation [7]. Hybrid CNN-Transformer models [8] have also been explored to balance local and global modeling, yet these approaches often rely on deep and complex extraction networks with large parameter counts and high training costs.

In summary, despite the progress of CNNs, Transformers, and hybrid models in colonoscopy image segmentation, existing approaches still struggle to balance model complexity, fine-grained detail fidelity, and semantic consistency, especially under challenging conditions such as low contrast, flat morphology, and blurred boundaries [9]. Moreover, deeper or hybrid

architectures often incur heavy computation and large parameter counts, limiting their feasibility in real-time clinical deployment.

Motivated by these challenges, we argue that an effective polyp segmentation framework must meet three requirements: (1) enhance the discriminability of deep features to suppress redundancy, (2) enforce semantic alignment between encoder and decoder to improve consistency, and (3) achieve these goals under lightweight design constraints to ensure clinical applicability. To this end, we propose the Hierarchical Contextual Attention Network (HCA-Net), which introduces specialized modules at both the encoder and decoder stages to jointly optimize global semantic modeling and local boundary recovery under strict efficiency constraints, requiring only 10.1 M parameters and 9.3 GFLOPs while achieving real-time inference at approximately 60 frames per second (FPS) for online polyp segmentation during colonoscopy. This capability enables timely and reliable computer-assisted support, improving the practical usability of automated polyp segmentation in routine clinical workflows.

The main contributions of this paper are summarized as follows:

- 1) We propose the HCA, a lightweight attention mechanism which balances global context modeling and local discriminability with reduced computational cost.
- 2) We design the RS-DPD module that alleviates deep feature redundancy while preserving fine-grained details.
- 3) We develop the BA-SAU module that enforces semantic alignment and enhances boundary sensitivity during decoding.
- 4) Extensive experiments demonstrate that redundancy suppression and cross-layer semantic alignment are particularly effective for segmenting small polyps and boundary-ambiguous regions. With a boundary-aware composite loss, the proposed method achieves SOTA performance on multiple public benchmarks with high efficiency.

II. RELATED WORK

Automatic polyp segmentation plays a crucial role in clinical practice, as it contributes to reducing cancer mortality. Traditional segmentation methods, such as thresholding, edge detection, region-based, and morphological approaches [10],[11], mainly rely on low-level features including texture, intensity, shape, and color. However, due to their dependence on manually designed criteria, these methods often fail to achieve satisfactory results in complex scenarios or in detecting diminutive polyps.

A. CNNs Based Approaches

In recent years, the rapid development of artificial intelligence has greatly advanced research on deep learning-based automated polyp segmentation, yielding remarkable achievements. CNNs have long served as the backbone of this field, giving rise to numerous classical architectures. Among them, U-Net is the most representative model. Its encoder-decoder structure and skip connections maintain architectural simplicity and computational efficiency, while significantly improving segmentation accuracy in medical imaging. Nevertheless, U-Net still exhibits limitations when handling highly complex inputs or diverse

targets. To address this, researchers have continuously optimized U-shaped architectures, leading to a variety of improved methods. For example, Zhou et al. proposed UNet++ [12], redesigned skip connections to minimize semantic gaps between features, thereby enhancing segmentation performance. Yeung et al. introduced Focus U-Net [13], which incorporates both spatial and channel-based attention into a unified Focus Gate module to encourage selective learning of polyp features. Along similar lines, ASD-Net [14] improves U-Net based segmentation through adaptive convolutional optimization and multi-scale context modeling, while DenLsNet [15] enhances convolutional feature extraction by incorporating sequence modeling. In addition, DEFIF-Net [16] adopts a lightweight dual-encoding architecture with feature interaction and channel reconstruction mechanisms to further improve segmentation accuracy while reducing model complexity. Notably, CaraNet [17] and PraNet [18] both leverage reverse attention modules to boost segmentation performance. Despite their success, CNN-based approaches suffer from intrinsic limitations in capturing global dependencies [19].

B. Transformer Based Approaches

Recently, the introduction of Transformers has opened a new pathway for medical image segmentation. Chen et al. proposed TransUNet [20], which combines the global modeling capability of Transformers with the fine localization ability of U-Net, effectively restoring local spatial information for more detailed segmentation. Furthermore, Vision Transformers (ViTs) [6], [21], with their strong global modeling ability, provide new technical support for polyp segmentation. In particular, methods based on Pyramid Vision Transformer (PVT) [22] leverage hierarchical feature extraction and multi-scale fusion to substantially improve segmentation accuracy and robustness. For example, Liu et al. proposed CAFE-Net [23], which introduces cross-layer feature supplementation, cross-attention, and multi-scale aggregation after the PVT encoder to bridge semantic gaps and enhance segmentation performance. Rahman et al. introduced CASCADE [24], which integrates attention gates with convolutional attention modules to fuse multi-scale features while suppressing background noise, and further accelerates convergence with multi-stage feature and loss aggregation. Wang et al. proposed MF-Net [25], combining PVT's global modeling with CNN's local feature extraction. Through multi-layer fusion, hybrid enhancement, and a pyramid prediction head, MF-Net enables efficient cross-scale feature interaction and fine boundary perception. Xu et al. introduced CIFFormer [26], which reconstructs the PVT framework via contextual information flow, integrating global-local feature collaboration with multi-density residual modules to improve fine-grained detail modeling and camouflaged polyp recognition.

C. CNNs-Transformer Hybrid Approaches

However, many Transformer-based approaches still face challenges in multi-layer feature interaction and fine-grained detail reconstruction. As a result, they often struggle to balance global semantic modeling with local detail representation when dealing with polyps that exhibit diverse morphologies, blurred

boundaries, or strong camouflage. To address this issue, hybrid networks that combine the global dependency modeling of Transformers with the local feature extraction and boundary representation strengths of CNNs have gradually become a major research trend. For instance, Zhang et al. proposed HSNet [27], which integrates Transformer and CNN architectures, introducing cross-semantic attention, hybrid semantic complementation, and multi-scale prediction modules to enhance the capture of morphological variations and fine details. Khan et al. proposed BiCoMA [28], a bilateral CNN-Transformer framework that integrates global contextual modeling and local spatial refinement through dual collaborative streams with multi-modal attention, achieving accurate and efficient polyp segmentation. Chu et al. proposed PFPRNet [29], a Transformer-based polyp segmentation framework that employs a phase-wise feature pyramid and a low-layer retention mechanism to enhance feature modeling. Similarly, Khan et al. proposed a Pyramidal Attention Mechanism (PAM) [30] with progressive multi-stage feature refinement to improve boundary adherence and structural discriminability. Li et al. further proposed Asps [31], which, based on the Segment Anything Model [32] (SAM), incorporates cross-branch feature enhancement and uncertainty-guided regularization to combine the advantages of CNNs and ViTs, thereby mitigating local detail loss and improving out-of-distribution generalization. Manan et al. proposed LWAH-Net [33], a lightweight CNN-Transformer hybrid network that integrates multi-scale feature enhancement, an attention bottleneck, and a hybrid loss design to achieve a balance between accuracy and efficiency in polyp segmentation. Pan et al. proposed MSBP-Net [34], which introduces improved receptive field blocks, boundary prediction modules, and shallow filtering blocks to achieve effective and low-complexity polyp segmentation.

Although encoder-decoder based architectures have driven remarkable progress in colorectal polyp segmentation, several critical challenges remain. First, balancing global semantic modeling with fine-grained detail reconstruction is essential for improving sensitivity to low-contrast and blurry-boundary polyps. Second, enhancing the discriminative ability to distinguish highly camouflaged polyps from surrounding tissues remains a non-trivial task. Third, achieving high segmentation accuracy while maintaining low parameter counts and computational efficiency is crucial for enabling real-time clinical deployment [35]. To address these challenges, we propose a lightweight segmentation framework, HCA-Net. By jointly modeling global semantics and local boundaries at both the encoding and decoding stages, the framework effectively mitigates the inherent limitations of conventional encoder-decoder paradigms, while achieving a superior balance among semantic consistency, boundary sensitivity, and computational efficiency.

III. METHODOLOGY

While conventional encoder-decoder architectures provide a reasonable balance between semantic abstraction and spatial recovery, they often suffer from deep feature redundancy, semantic misalignment, and insufficient boundary restoration. Building on these observations, we design HCA-Net with the goal of enhancing discriminability and consistency under lightweight

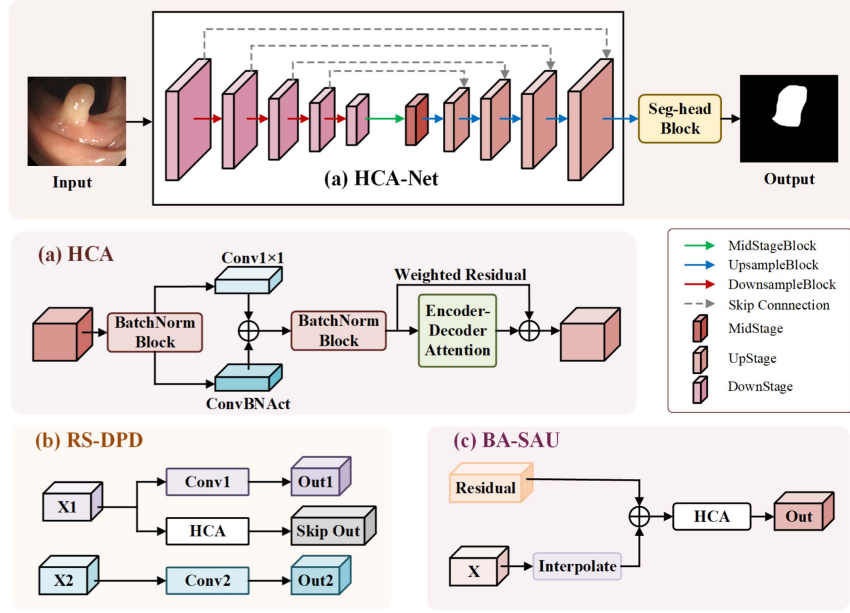


Fig. 2. The overall architecture of the HCA-Net.

constraints. As shown in the Fig. 2, the framework integrates two complementary modules: RS-DPD in the encoder, which reduces redundancy while preserving fine details, and BA-SAU in the decoder, which refines semantic alignment and boundary recovery. Both modules are underpinned by the proposed HCA, enabling efficient yet effective feature modeling across stages.

A. Hierarchical Contextual Attention

Traditional self-attention mechanisms compute dense pairwise dependencies over full-resolution feature maps, resulting in quadratic complexity with respect to spatial size. Such designs are computationally prohibitive for high-resolution medical images. In contrast, conventional CNN-based methods rely on local receptive fields and often fail to capture long-range contextual dependencies, which leads to semantic inconsistency under challenging conditions such as low contrast or blurred boundaries.

To address this trade-off, we design Hierarchical Contextual Attention (HCA), a lightweight attention mechanism that combines convolutional projection (to preserve local discriminability) with adaptive pooling (to compress spatial resolution for global context modeling). This achieves efficient feature interaction while significantly reducing computation. As illustrated in Fig. 2(a), HCA is instantiated as a structured attention block, in which the Encoder-Decoder Attention module constitutes a core component and is consistently applied across both encoding and decoding stages.

Given an input feature map $\mathbf{F} \in \mathbb{R}^{H \times W \times C}$, HCA first performs feature normalization followed by a dual-branch local projection. One branch employs a 1×1 convolution to re-embed channel-wise information with minimal computational overhead, while the other branch applies a 3×3 convolution with batch normalization and activation (ConvBNAct) to enhance local spatial context. The outputs of these two branches are fused

via element-wise summation, enabling complementary modeling of channel-wise abstraction and spatial detail refinement:

$$\mathbf{F}_{loc} = \phi_{1 \times 1}(\mathbf{F}) + \phi_{3 \times 3}(\mathbf{F}). \quad (1)$$

The fused local features are further normalized and passed to an Encoder-Decoder Attention module, which implements hierarchical contextual attention by modeling global dependencies in a compressed spatial domain, as illustrated in Fig. 3. Specifically, query, key, and value features are generated through 1×1 convolutional projections and adaptive average pooling:

$$\begin{cases} \mathbf{Q} = \phi_q(\text{Pool}(\mathbf{F}_{loc})), \\ \mathbf{K} = \phi_k(\text{Pool}(\mathbf{F}_{loc})), \\ \mathbf{V} = \phi_v(\text{Pool}(\mathbf{F}_{loc})). \end{cases} \quad (2)$$

Here, $\text{Pool}(\cdot)$ reduces the spatial resolution from $H \times W$ to $\frac{H}{s} \times \frac{W}{s}$, where the scale factor s is fixed across all stages to balance computational efficiency and contextual coverage. In our implementation, all convolutional projections $\phi_q(\cdot)$, $\phi_k(\cdot)$, $\phi_v(\cdot)$, and $\phi_z(\cdot)$ are realized using 1×1 convolutions, enabling efficient channel re-embedding without introducing additional spatial overhead.

Self-attention is computed in the compressed feature space:

$$\mathbf{A} = \text{Softmax}\left(\frac{\mathbf{Q}\mathbf{K}^\top}{\sqrt{d}}\right), \quad \mathbf{U} = \mathbf{A}\mathbf{V}, \quad (3)$$

where the embedding dimension d for the query and key projections is set to a small constant, which is sufficient to capture coarse global dependencies while significantly reducing computational complexity.

The resulting contextual features are then upsampled and projected back to the original channel space via a 1×1 convolution:

$$\mathbf{F}_{att} = \phi_z(\text{Upsample}(\mathbf{U})), \quad (4)$$

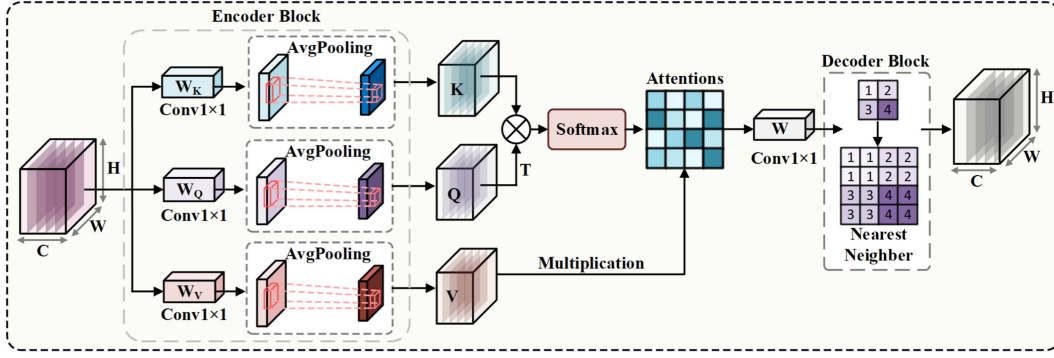


Fig. 3. Overall architecture of the proposed Encoder-Decoder Attention Block.

where $\phi_z(\cdot)$ denotes a 1×1 convolution, and the upsampling operation is implemented using nearest-neighbor interpolation.

Finally, HCA adopts a weighted residual connection to inject global contextual information into the locally refined features:

$$\mathbf{F}_{out} = \mathbf{F}_{loc} + \alpha \cdot \mathbf{F}_{att}, \quad (5)$$

where α is a small learnable or fixed scaling factor that stabilizes optimization and prevents excessive distortion of fine-grained structural details.

By computing attention in a pooled low-resolution space and injecting the resulting contextual features via upsampling and weighted residual refinement, HCA avoids dense token-level interactions and achieves an effective trade-off between lightweight design and modeling capability. This hierarchical design combines dual-branch local feature enhancement with pooled global attention, enabling HCA to capture long-range semantic dependencies while preserving boundary-sensitive local representations, making it well suited for encoder-decoder architectures in polyp segmentation. It also serves as a unified attention kernel, reused in both encoder and decoder modules.

B. Redundancy-Suppressed Dual-Path Downsampling

In standard convolution-pooling downsampling, deep features often suffer from channel redundancy and loss of fine-grained information. Although skip connections partially restore details, they cannot fully compensate for redundancy-induced degradation. We propose Redundancy-Suppressed Dual-Path Downsampling (**RS-DPD**), which introduces a dual-path strategy: (1) a semantic-dominant path enhanced by HCA to extract low-redundancy features, and (2) a structure-preserving path to retain boundary and texture cues. Their fusion achieves complementary representation of semantics and details.

Formally, given input $\mathbf{F}$:

$$\mathbf{F}_{main} = \text{DownConv}(\text{HCA}(\mathbf{F})), \quad \mathbf{F}_{aux} = \text{DownConv}(\mathbf{F}), \quad (6)$$

$$\mathbf{F}_{dpd} = \mathbf{F}_{main} + \mathbf{F}_{aux}, \quad \mathbf{F}_{skip} = \text{HCA}(\mathbf{F}). \quad (7)$$

Unlike traditional downsampling that sacrifices either semantic abstraction or structural fidelity, RS-DPD achieves redundancy suppression and detail preservation simultaneously,

providing semantically discriminative and structurally faithful representations.

C. Boundary-Aware Semantic Alignment Upsampling

During decoding, spatial resolution is restored but semantic misalignment and insufficient boundary recovery often occur. Direct fusion of encoder and decoder features can introduce noise, as their semantic levels are inconsistent. We propose Boundary-Aware Semantic Alignment Upsampling (**BA-SAU**), which adapts HCA into a cross-attention form. Decoder features act as queries, while encoder skip features act as keys and values. This design enforces semantic alignment while enhancing boundary sensitivity.

Let $\mathbf{F}_d$ be decoder features and $\mathbf{F}_e$ encoder features:

$$\mathbf{F}_{sau} = \mathbf{F}_d + \beta \cdot \text{HCA}(\mathbf{F}_d, \mathbf{F}_e), \quad (8)$$

where β is a learnable coefficient.

Different from conventional skip connections, BA-SAU introduces boundary-aware cross-attention, ensuring semantic consistency across scales and enhancing boundary recovery in challenging polyp cases.

In summary, HCA serves as the lightweight attention kernel, RS-DPD alleviates deep feature redundancy while preserving details in the encoder, and BA-SAU enforces semantic alignment and boundary refinement in the decoder. Together, these modules form the HCA-Net framework, a principled hierarchical encoder-decoder attention design tailored for accurate and efficient polyp segmentation.

D. Loss Function

To effectively optimize the proposed HCA-Net and enhance its segmentation performance in low-contrast and boundary-ambiguous regions, we design a composite loss function that combines weighted binary cross-entropy (weighted BCE, $\mathcal{L}_{BCE}^w$) and weighted intersection-over-union (weighted IoU, $\mathcal{L}_{IoU}^w$), with deep supervision applied at multiple scales.

1) **Boundary-Aware Weighting:** Let $\mathbf{G} \in \{0, 1\}^{H \times W}$ denote the ground-truth mask and $\mathbf{P} \in [0, 1]^{H \times W}$ the predicted probability map. To emphasize structurally uncertain regions, particularly polyp boundaries, an importance coefficient α_{ij} is assigned to each pixel (i, j) , computed using gradient operators

(e.g., Sobel) combined with local smoothing. The final weight is defined as:

$$w_{ij} = 1 + \kappa \alpha_{ij}, \quad \kappa > 0, \quad (9)$$

where κ controls the amplification of boundary regions.

2) **Weighted Binary Cross-Entropy**: The weighted BCE term is formulated as:

$$\mathcal{L}_{\text{BCE}}^w = - \frac{1}{\sum_{i=1}^H \sum_{j=1}^W w_{ij} + \varepsilon} \sum_{i=1}^H \sum_{j=1}^W w_{ij} (g_{ij} \log(p_{ij} + \varepsilon) + (1 - g_{ij}) \log(1 - p_{ij} + \varepsilon)). \quad (10)$$

where g_{ij} and p_{ij} represent the ground-truth label and predicted probability for pixel (i, j) , respectively, and ε is a small constant for numerical stability.

3) **Weighted IoU**: Structural consistency is modeled by the weighted IoU term:

$$\mathcal{L}_{\text{IoU}}^w = 1 - \frac{\sum_{i=1}^H \sum_{j=1}^W w_{ij} g_{ij} p_{ij}}{\sum_{i=1}^H \sum_{j=1}^W w_{ij} (g_{ij} + p_{ij} - g_{ij} p_{ij}) + \varepsilon}. \quad (11)$$

4) **Multi-Scale Deep Supervision**: Since HCA-Net adopts a hierarchical encoder-decoder architecture with predictions at multiple stages, deep supervision is applied at each scale. For the s -th predicted map $\mathbf{P}^{(s)}$, the loss is defined as:

$$\mathcal{L}^{(s)} = \mathcal{L}_{\text{BCE}}^w(\mathbf{P}^{(s)}, \mathbf{G}) + \mathcal{L}_{\text{IoU}}^w(\mathbf{P}^{(s)}, \mathbf{G}). \quad (12)$$

The total loss is then the weighted sum over all supervised stages:

$$\mathcal{L}_{\text{total}} = \sum_{s \in \mathcal{S}} \omega_s \mathcal{L}^{(s)}, \quad (13)$$

where $\mathcal{S}$ denotes the set of supervised stages and ω_s is a stage-specific weighting factor. In practice, higher-resolution outputs are assigned larger weights to ensure fine-grained prediction accuracy.

This composite loss enforces pixel-level accuracy through the weighted BCE and structural consistency at the region level via the weighted IoU. The boundary-aware weighting enhances sensitivity to ambiguous regions, while the multi-scale deep supervision accelerates convergence and mitigates semantic degradation during training.

IV. EXPERIMENT

In this section, we describe the experimental configurations to ensure the reproducibility and reliability of our results. Specifically, we introduce the datasets and their preprocessing procedures, the evaluation metrics, the key hyperparameters used during training, and the hardware/software environment. Based on these settings, we report the performance of our model on multiple public datasets and compare it with SOTA methods. Furthermore, ablation studies are conducted to analyze the contribution of each module and validate their effectiveness in improving overall performance.

A. Datasets

In order to assess the performance of the proposed polyp segmentation model, we conducted experiments using five publicly

TABLE I
DETAILS OF THE POLYP SEGMENTATION DATASETS USED IN OUR EXPERIMENTS

Datasets	Images	Input sizes	Train	Test
Kvasir-SEG	1000	352 × 352	900	100
CVC-ClinicDB	612	352 × 352	551	61
CVC-ColonDB	380	352 × 352	–	380
CVC-300	60	352 × 352	–	60
ETIS-LaribPolypDB	196	352 × 352	–	196

available polyp datasets, which have been widely adopted in prior SOTA methods for benchmarking. A detailed description of these datasets is provided below:

- 1) **Kvasir-SEG [36]**: The Kvasir-SEG dataset contains 1,000 polyp images of varying shapes and sizes, along with corresponding mask images. The image resolutions range from 332×487 to 1920×1072 pixels. This dataset is suitable for training and evaluating polyp detection and segmentation algorithms.
- 2) **ClinicDB [37]**: The ClinicDB dataset consists of 612 polyp images derived from 31 colonoscopy sequences, with a resolution of 384×288 pixels. Each image is annotated by a medical expert and includes a corresponding polyp mask. This dataset provides high-quality annotations for polyp image analysis.
- 3) **ColonDB [38]**: The ColonDB dataset comprises 380 polyp images extracted from 15 colonoscopy videos, each showing a single polyp. The image resolution is 574×500 pixels, making it suitable for training segmentation models for single polyps.
- 4) **CVC-300 [39]**: The CVC-300 dataset contains 60 polyp images, each with a resolution of 574×500 pixels. This dataset is useful for developing and evaluating small sample polyp segmentation models.
- 5) **ETIS-LaribpolypDB [40]**: The ETIS-LaribpolypDB dataset includes 196 polyp images from 44 different polyps, with a resolution of 1225×966 pixels, covering 34 colonoscopy sequences. This dataset is suitable for training deep learning algorithms and researching polyp detection techniques.

We adopt the standard training and testing set division strategy commonly used in previous works, as implemented by PraNet [18], across five datasets: Kvasir-SEG, CVC-ClinicDB, CVC-ColonDB, CVC-300, and ETIS-LaribPolypDB. Specifically, 90% of the Kvasir-SEG and CVC-ClinicDB datasets are allocated for training, while the remaining 10% are reserved for testing. Additionally, we assess the models' ability to generalize by testing them on unseen datasets, including CVC-ColonDB, CVC-300, and ETIS-LaribPolypDB. The details of the polyp segmentation datasets used in our experiments are shown in Table I.

B. Evaluation Metrics

To comprehensively evaluate the predictive performance of the proposed model, we adopt a set of metrics widely used in both semantic segmentation and salient object detection. Specifically, we first employ the **mean Intersection over Union (mIoU)** and **mean Dice coefficient (mDice)** as the fundamental

segmentation metrics, which measure the spatial overlap between predictions and ground-truth annotations. In addition, to further assess the model's capability in saliency preservation and boundary recovery, we incorporate four complementary metrics commonly used in the Salient Object Detection (SOD) field: **structure-measure** (S_α) [41], **weighted F-measure** (F_β^ω) [42], **enhanced alignment measure** (E_ϕ) [43], **mean absolute error** (MAE) [44] and **Boundary F1 (BF1)** [45]. These metrics provide a multi-perspective quantitative evaluation, ensuring a fair and comprehensive assessment of segmentation performance.

a) mIoU and mDice: mIoU and mDice are two standard metrics for semantic segmentation, quantifying the spatial consistency between the predicted mask $\mathbf{P}$ and the ground truth $\mathbf{G}$. They are defined as:

$$\text{IoU} = \frac{|\mathbf{P} \cap \mathbf{G}|}{|\mathbf{P} \cup \mathbf{G}|}, \quad \text{Dice} = \frac{2|\mathbf{P} \cap \mathbf{G}|}{|\mathbf{P}| + |\mathbf{G}|}. \quad (14)$$

For multi-class segmentation tasks, results are averaged across all categories to obtain mIoU and mDice. Since this work focuses on single-class polyp segmentation, the above definitions are directly applicable.

b) S_α : The structure-measure evaluates the similarity between prediction and ground truth in terms of both object-aware structure and region-aware structure, defined as a weighted combination:

$$S_\alpha = \alpha S_o + (1 - \alpha) S_r, \quad (15)$$

where S_o and S_r denote object-aware and region-aware similarities, respectively, and $\alpha \in [0, 1]$ is a balance parameter.

c) F_β^ω : Traditional F-measure may be biased under imbalanced distributions of foreground and background. The weighted version F_β^ω mitigates this by incorporating spatial weighting:

$$F_\beta^\omega = \frac{(1 + \beta^2) \cdot \text{Precision}^\omega \cdot \text{Recall}^\omega}{\beta^2 \cdot \text{Precision}^\omega + \text{Recall}^\omega}, \quad (16)$$

where $\beta^2 = 0.3$ emphasizes precision, and $\text{Precision}^\omega, \text{Recall}^\omega$ are the weighted precision and recall.

d) E_ϕ : The enhanced alignment measure establishes a unified evaluation framework between pixel- and image-level similarities, defined as:

$$E_\phi = \frac{1}{W \times H} \sum_{x=1}^W \sum_{y=1}^H \phi(\mathbf{P}(x, y), \mathbf{G}(x, y)), \quad (17)$$

where $\phi(\cdot)$ is the enhanced consistency function, and W, H denote the width and height of the image.

e) MAE: MAE measures the pixel-wise difference between the predicted probability map and the ground truth:

$$\text{MAE} = \frac{1}{W \times H} \sum_{x=1}^W \sum_{y=1}^H |\mathbf{P}(x, y) - \mathbf{G}(x, y)|. \quad (18)$$

It directly reflects the overall prediction error magnitude.

f) Boundary F1: Boundary F1 (BF1) measures the alignment accuracy between predicted and ground-truth boundaries. Boundary precision and recall are computed based on boundary pixel matching within a tolerance. The Boundary F1 score is

then defined as:

$$\text{BF1} = \frac{2 \cdot \text{Precision}_b \cdot \text{Recall}_b}{\text{Precision}_b + \text{Recall}_b}, \quad (19)$$

where Precision_b and Recall_b denote the boundary-based precision and recall, respectively. A higher BF1 score indicates better boundary localization and shape consistency.

C. EXPERIMENT DETAILS

The proposed model is implemented in PyTorch 2.5.1 and trained on an Ubuntu 22.04 system equipped with an NVIDIA GeForce RTX 4090 GPU. The software environment includes CUDA 12.4 and Python 3.12. During training, the Adam optimizer is employed to update model parameters, with the initial learning rate set to 1×10^{-4} . A polynomial decay schedule is adopted for dynamic adjustment, such that at the t -th iteration the learning rate is defined as:

$$\eta_t = \eta_0 \left(1 - \frac{t}{T}\right)^2, \quad (20)$$

where η_0 denotes the initial learning rate and T is the total number of iterations.

The model is trained for 200 epochs with a batch size of 16, and all input images are resized to a fixed resolution of 352×352 . To improve generalization capability, various data augmentation techniques are applied during training, including random flipping, rotation, cropping, as well as random adjustments of contrast, brightness, and sharpness. All experiments are repeated five times with different random seeds, and the mean and standard deviation are reported. Statistical significance is evaluated using paired t-tests with a significance level of $p < 0.05$.

D. QUANTITATIVE RESULTS

To evaluate the effectiveness of our proposed model, we conducted comprehensive comparisons between HCA-Net and several representative SOTA approaches on five public colorectal polyp segmentation benchmark datasets, including U-Net [3], U-Net++ [12], PraNet [18], CaraNet [17], Polyp-PVT [21], HSNet [27], PVT-CASCADE [24], CAFE-Net [23], Polyp-LVT [35], MF-Net [25], PFPRNet [29], MSBP-Net [34], and CFFormer [26].

On the Kvasir-SEG dataset (Table II), HCA-Net achieves mDice and mIoU scores of 0.936 and 0.897, respectively, surpassing the second-best method CFFormer (0.926 and 0.876) by 1.1% and 2.4%. On the CVC-ClinicDB dataset (Table III), HCA-Net obtains 0.953 in mDice and 0.909 in mIoU, yielding improvements of 0.8% and 1.1% over CAFE-Net (0.943 and 0.899). On the more challenging CVC-ColonDB dataset (Table IV), HCA-Net reaches 0.834 in mDice and 0.769 in mIoU, outperforming MF-Net (0.825 and 0.751) by 1.1% and 2.4%. On the CVC-300 dataset (Table V), HCA-Net records 0.912 in mDice and 0.873 in mIoU, achieving relative gains of 1.0% and 1.6% compared to MSBP-Net (0.903 and 0.859). Finally, on the most challenging ETIS-LaribpolypDB dataset (Table VI), HCA-Net attains 0.833 in mDice and 0.789 in mIoU,

TABLE II

EXPERIMENTAL COMPARISON OF DIFFERENT MODELS ON KVASIR-SEG

Methods	Year	mDice	mIoU	F_{β}^w	S_{α}	mE_{ξ}	MAE	BF1
U-Net	2015	0.806	0.718	0.784	0.842	0.890	0.052	84.54
U-Net++	2018	0.797	0.704	0.767	0.827	0.884	0.056	82.37
PraNet	2020	0.897	0.845	0.889	0.916	0.945	0.026	85.75
CaraNet	2021	0.911	0.861	0.902	0.924	0.953	0.024	86.16
Polyp-PVT	2021	0.920	0.871	0.915	0.927	0.962	0.024	89.56
HSNet	2022	0.928	0.884	0.924	0.933	0.963	0.017	90.45
PVT-CASCADE	2023	0.924	0.875	0.920	0.929	0.960	0.022	91.26
CAFE-Net	2024	0.933	0.889	0.927	0.939	0.967	0.019	91.98
Polyp-LVT	2024	0.909	0.851	0.913	0.922	0.941	0.024	89.44
MF-Net	2025	0.915	0.863	0.939	0.924	0.962	0.022	90.70
MSBP-Net	2025	0.919	0.868	0.916	0.926	0.961	0.023	90.47
CIFFormer	2025	0.926	0.876	0.929	0.939	0.967	0.019	91.08
PFPRNet	2025	0.930	0.881	0.920	0.935	0.971	0.019	91.82
HCA-Net (Ours)	-	0.936	0.897	0.930	0.941	0.966	0.018	91.74

TABLE III

EXPERIMENTAL COMPARISON ON CVC-CLINICDB

Methods	Year	mDice	mIoU	F_{β}^w	S_{α}	mE_{ξ}	MAE	BF1
U-Net	2015	0.830	0.769	0.829	0.887	0.918	0.020	85.54
U-Net++	2018	0.838	0.777	0.837	0.895	0.920	0.019	85.83
PraNet	2020	0.916	0.870	0.915	0.946	0.973	0.007	87.24
CaraNet	2021	0.936	0.887	0.934	0.953	0.985	0.006	90.48
Polyp-PVT	2021	0.939	0.892	0.939	0.948	0.987	0.006	92.36
HSNet	2022	0.936	0.890	0.934	0.949	0.982	0.006	92.84
PVT-CASCADE	2023	0.940	0.895	0.940	0.952	0.983	0.006	92.39
CAFE-Net	2024	0.943	0.899	0.941	0.957	0.986	0.006	92.58
Polyp-LVT	2024	0.935	0.882	0.938	0.934	0.982	0.007	92.85
MF-Net	2025	0.931	0.884	0.953	0.949	0.983	0.006	93.03
MSBP-Net	2025	0.940	0.892	0.927	0.939	0.976	0.007	92.81
CIFFormer	2025	0.944	0.897	0.941	0.934	0.972	0.007	93.19
PFPRNet	2025	0.949	0.903	0.947	0.957	0.989	0.006	93.16
HCA-Net (Ours)	-	0.953	0.909	0.947	0.939	0.981	0.006	93.72

TABLE IV

EXPERIMENTAL COMPARISON ON CVC-COLONDB

Methods	Year	mDice	mIoU	F_{β}^w	S_{α}	mE_{ξ}	MAE	BF1
U-Net	2015	0.551	0.467	0.532	0.721	0.726	0.051	65.64
U-Net++	2018	0.498	0.410	0.478	0.686	0.719	0.057	69.62
PraNet	2020	0.703	0.634	0.695	0.815	0.835	0.030	81.35
CaraNet	2021	0.755	0.682	0.740	0.843	0.872	0.026	81.07
Polyp-PVT	2021	0.791	0.711	0.779	0.855	0.900	0.019	82.59
HSNet	2022	0.802	0.724	0.784	0.864	0.902	0.015	81.66
PVT-CASCADE	2023	0.801	0.725	0.787	0.862	0.903	0.029	82.17
CAFE-Net	2024	0.820	0.740	0.803	0.874	0.918	0.026	82.17
Polyp-LVT	2024	0.815	0.733	0.819	0.844	0.902	0.027	79.63
MF-Net	2025	0.825	0.751	0.864	0.887	0.929	0.024	81.28
MSBP-Net	2025	0.810	0.731	0.810	0.861	0.912	0.029	82.19
CIFFormer	2025	0.823	0.741	0.851	0.874	0.918	0.026	82.74
PFPRNet	2025	0.819	0.741	0.843	0.882	0.915	0.028	82.43
HCA-Net (Ours)	-	0.834	0.769	0.887	0.899	0.927	0.020	82.87

showing notable improvements of 1.3% and 6.9% over CAFE-Net (0.822 and 0.738). Overall, these results demonstrate that HCA-Net consistently outperforms SOTA approaches across five public benchmarks, with particularly significant advantages on small-scale and complex datasets, highlighting its robustness and generalization capability.

To further assess whether the observed improvements are attributable to systematic advantages rather than random fluctuations, we evaluate all methods across multiple datasets and report

TABLE V

EXPERIMENTAL COMPARISON ON CVC-300

Methods	Year	mDice	mIoU	F_{β}^w	S_{α}	E_{ϕ}	MAE	BF1
U-Net	2015	0.710	0.627	0.684	0.843	0.876	0.022	72.03
U-Net++	2018	0.707	0.624	0.687	0.839	0.898	0.018	74.74
PraNet	2020	0.871	0.797	0.843	0.925	0.972	0.010	86.06
CaraNet	2021	0.880	0.811	0.856	0.869	0.963	0.020	87.20
Polyp-PVT	2021	0.887	0.821	0.898	0.965	0.873	0.016	86.79
HSNet	2022	0.891	0.823	0.858	0.871	0.965	0.014	87.28
PVT-CASCADE	2023	0.881	0.824	0.867	0.882	0.957	0.013	86.76
CAFE-Net	2024	0.882	0.820	0.870	0.901	0.955	0.012	87.69
Polyp-LVT	2024	0.888	0.816	0.911	0.929	0.953	0.010	85.57
MF-Net	2025	0.897	0.836	0.918	0.941	0.966	0.007	88.05
MSBP-Net	2025	0.903	0.859	0.927	0.942	0.967	0.019	87.23
CIFFormer	2025	0.883	0.822	0.912	0.939	0.961	0.010	87.87
PFPRNet	2025	0.891	0.833	0.917	0.928	0.950	0.013	87.95
HCA-Net (Ours)	-	0.912	0.873	0.924	0.943	0.966	0.011	88.80

TABLE VI

EXPERIMENTAL COMPARISON ON ETIS-LARIBPOLYPDB

Methods	Year	mDice	mIoU	F_{β}^w	S_{α}	E_{ϕ}	MAE	BF1
U-Net	2015	0.324	0.254	0.287	0.614	0.626	0.044	50.24
U-Net++	2018	0.344	0.267	0.308	0.626	0.643	0.039	48.16
PraNet	2020	0.656	0.590	0.631	0.810	0.819	0.017	72.58
CaraNet	2021	0.756	0.681	0.713	0.856	0.869	0.020	73.01
Polyp-PVT	2021	0.743	0.667	0.710	0.846	0.874	0.014	73.58
HSNet	2022	0.800	0.717	0.753	0.877	0.909	0.014	73.73
PVT-CASCADE	2023	0.795	0.707	0.757	0.881	0.892	0.018	72.57
CAFE-Net	2024	0.822	0.738	0.775	0.898	0.917	0.014	74.60
Polyp-LVT	2024	0.816	0.736	0.768	0.875	0.910	0.016	74.61
MF-Net	2025	0.821	0.750	0.850	0.901	0.925	0.011	75.45
MSBP-Net	2025	0.795	0.718	0.770	0.873	0.899	0.019	74.62
CIFFormer	2025	0.810	0.746	0.858	0.889	0.918	0.013	75.11
PFPRNet	2025	0.828	0.764	0.839	0.908	0.935	0.012	74.96
HCA-Net (Ours)	-	0.833	0.789	0.867	0.919	0.920	0.012	75.52

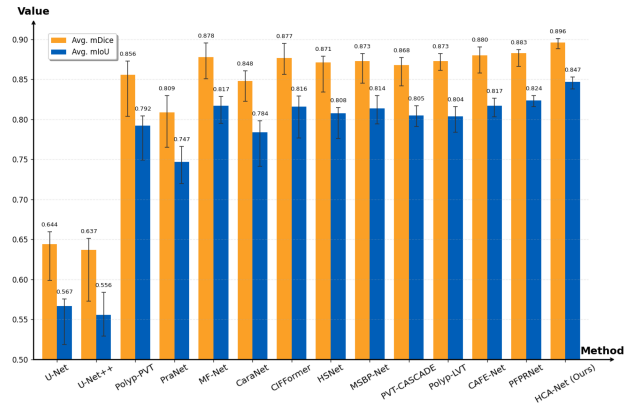


Fig. 4. Average mDice and mIoU of different polyp segmentation methods across five public datasets.

the mean performance together with cross-dataset variability. Fig. 4 summarizes the average segmentation performance of different methods in terms of mDice and mIoU across five public polyp segmentation datasets, with error bars indicating the variability of performance among datasets. As observed in the figure, conventional CNN-based baselines (e.g., U-Net and U-Net++) not only yield relatively low average performance but also exhibit larger variability, suggesting limited robustness when exposed to heterogeneous datasets. Dedicated polyp segmentation methods such as PraNet and CaraNet achieve notable improvements, while Transformer-based architectures further

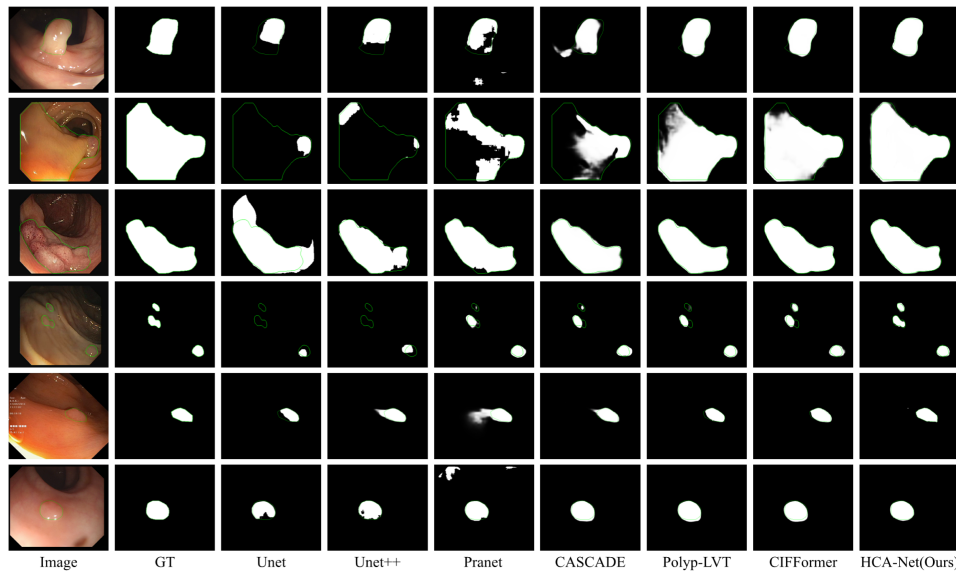


Fig. 5. Visualization of qualitative analysis of polyp segmentation results, where the green contours indicate the ground-truth boundaries.

enhance average performance by incorporating global contextual modeling. Importantly, the proposed HCA-Net achieves the highest average mDice and mIoU among all compared methods, while simultaneously exhibiting comparatively smaller error bars. Specifically, HCA-Net attains an average mDice of 0.8936 and an average mIoU of 0.8474, outperforming strong competitors including PFPRNet, CAFE-Net, and CIFFormer. The reduced variability across datasets indicates that the performance gains of HCA-Net are consistently maintained rather than driven by favorable outcomes on individual benchmarks, highlighting its robustness and practical applicability in real-world clinical scenarios.

E. QUALITATIVE RESULTS

As illustrated in Fig. 5, the qualitative comparison highlights the superiority of the proposed method over several SOTA approaches across diverse scenarios. In the first three rows, which depict medium-to-large polyps, HCA-Net is able to accurately delineate the overall lesion contours and fully cover the polyp regions, while other methods often produce incomplete boundaries or overly smoothed results. Even in cases with indistinct boundaries or complex lesion morphology, HCA-Net maintains more consistent and precise segmentation. In contrast, methods such as PraNet and Polyp-LVT show noticeable deficiencies in capturing the full extent of large polyps. In the fourth to sixth rows, which correspond to small polyps, HCA-Net demonstrates remarkable robustness. Whether facing low-contrast small targets or multiple polyps within the same scene, it precisely detects and separates each lesion, effectively avoiding the issues of missed detections, adhesion, or background mis-segmentation observed in other approaches. Particularly in the sixth row with multiple lesions, HCA-Net is capable of clearly distinguishing individual polyps from surrounding normal tissues.

We further analyze small polyp objects on four public datasets (CVC-ClinicDB, CVC-ColonDB, CVC-300, and ETIS-LaribPolypDB) to evaluate the performance of HCA-Net against

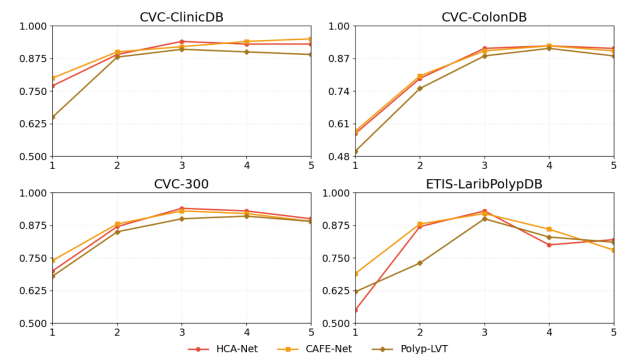


Fig. 6. Performance vs. size on the five polyp datasets. The x-axis denotes the proportion (%) of polyp size, and the y-axis represents the averaged mDice coefficient.

various SOTA models. As shown in Fig. 6, the overall trend indicates that the performance of all methods improves as polyp size increases, but the differences are most pronounced in the small-size range (1%–2%). HCA-Net outperforms CAFE-Net and Polyp-LVT in small polyp segmentation and maintains leading or comparable performance in the medium-size range. For large polyps, the performance of all methods becomes more stable and the gap narrows. Notably, on more challenging datasets such as CVC-ColonDB and ETIS-LaribPolypDB, HCA-Net demonstrates greater robustness in segmenting small objects and polyps with complex boundaries.

The Fig. 7 presents a comparison between the proposed HCA-Net and representative SOTA methods (CASCADE, Polyp-LVT, and CIFFormer) on local segmentation regions. To better highlight boundary differences, selected areas are enlarged. It can be observed that CASCADE and Polyp-LVT often suffer from boundary shifts or irregular contours, while CIFFormer produces relatively complete results but still exhibits over-smoothed edges or missing fine details. In contrast, HCA-Net achieves the most accurate boundary alignment, with predictions closely matching the ground truth. It effectively preserves contour

TABLE VII
ABLATION STUDY ON BACKBONE WIDTH (BASE_CHANNEL) AND THE COMPUTATIONAL OVERHEAD OF HCA

base_channel	HCA	CVC-ClinicDB		ETIS-LaribPolypDB		Param (MB)	GFLOPs (G)
		mDice	mIoU	mDice	mIoU		
7	–	0.801	0.748	0.662	0.627	3.45	2.15
7	+	0.902	0.847	0.792	0.742	5.12	3.39
10	–	0.822	0.773	0.693	0.642	6.82	4.05
10	+	0.928	0.879	0.812	0.763	8.94	6.01
12	–	0.843	0.797	0.721	0.672	7.92	6.21
12	+	0.953	0.909	0.833	0.790	10.13	9.33
14	–	0.846	0.802	0.724	0.676	10.77	8.43
14	+	0.956	0.911	0.837	0.794	13.68	12.09
17	–	0.851	0.805	0.729	0.681	15.91	12.11
17	+	0.958	0.913	0.840	0.799	20.36	17.98

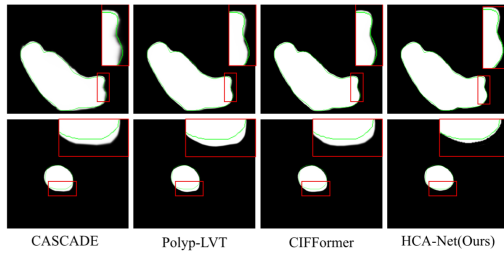


Fig. 7. Qualitative comparison highlighting fine-grained boundary modeling in polyp segmentation.

continuity and completeness, particularly in small polyps and complex boundary regions, thereby avoiding boundary blurring and over-expansion. These results demonstrate the superiority of HCA-Net in fine-grained structural modeling and boundary enhancement.

To further investigate the effectiveness of different modules in the proposed HCA-Net, we visualized the intermediate feature maps at each stage, as shown in Fig. 8. Taking the fourth row as an example, Down_stage1, Down_stage3, and Down_stage5 represent the feature outputs of the encoder at different depths. It can be observed that as the network goes deeper, the model gradually enhances its ability to capture semantic information. In particular, Down_stage5 exhibits strong responses in polyp regions, although some noise and incomplete localization remain.

In the decoding process, Up_stage5, Up_stage3, and Up_stage1 progressively integrate multi-scale features with skip connections. This fusion helps the model restore fine-grained details while maintaining global context. For example, Up_stage3 clearly strengthens the responses around polyp contours, effectively suppressing background interference. Out_stage further refines these predictions, producing highly consistent boundaries that closely match the ground truth.

Finally, the Prediction column demonstrates that the proposed HCA-Net can achieve segmentation results highly aligned with GT masks. Compared to earlier stages, the final predictions show accurate localization and smooth boundaries, particularly for polyps with blurred edges or small scales. These results validate that the encoder–decoder attention and dual-path downsampling design significantly enhance the model’s capacity for precise feature representation and boundary refinement.

Overall, these qualitative results indicate that HCA-Net achieves superior segmentation performance across varying polyp sizes, indistinct boundaries, and multi-lesion scenarios, thereby validating its effectiveness and robustness in diverse clinical settings.

F. ABLATION STUDY

1) Ablation Study on Backbone Width and HCA

To investigate the impact of backbone capacity and the computational overhead introduced by HCA, we conduct an ablation study by varying the backbone width (base_channel) from 7 to 17 and enabling or disabling HCA. As shown in Table VII, increasing the backbone width consistently improves baseline segmentation performance, but at the cost of substantially increased parameter size and GFLOPs, revealing a clear efficiency–capacity trade-off. Notably, introducing HCA leads to significant performance gains across all backbone widths, with more pronounced improvements observed for narrower backbones (base_channel=7 and 10), indicating that HCA effectively compensates for limited representational capacity through efficient global contextual modeling. While wider backbones (base_channel=14 and 17) also benefit from HCA, the marginal accuracy gains become progressively smaller relative to the rapidly growing computational overhead. In particular, the base_channel=12 configuration combined with HCA already achieves strong and stable performance on both CVC-ClinicDB and ETIS-LaribPolypDB, while maintaining a considerably lower model size and computational cost compared with wider settings. Therefore, base_channel=12 provides a favorable balance between segmentation accuracy and efficiency and is adopted as the default configuration in HCA-Net.

2) Ablation Study of Individual Modules

To further elucidate the individual contributions of each proposed component and their synergistic effects, Table VIII presents a systematic ablation study to analyze the independent and combined impacts of the HCA, RS-DPD, and BA-SAU modules on both the in-domain CVC-ClinicDB dataset and the cross-domain ETIS-LaribPolypDB dataset. Starting from the U-Net backbone, introducing HCA alone consistently improves segmentation performance, indicating that lightweight global

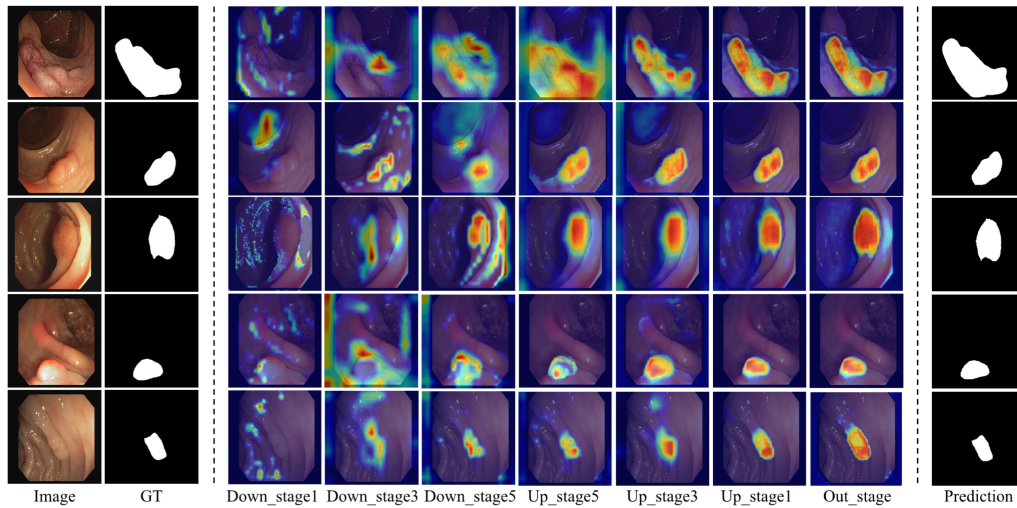


Fig. 8. Module visualization feature map.

TABLE VIII

SYSTEMATIC ABLATION STUDY OF THE PROPOSED MODULES ON CVC-CLINICDB (IN-DOMAIN) AND ETIS-LARIBPOLYPDB

Models	CVC-ClinicDB		ETIS-LaribPolypDB	
	mDice	mIoU	mDice	mIoU
Backbone	0.830	0.769	0.324	0.254
+HCA	0.886	0.812	0.621	0.564
+RS-DPD	0.914	0.839	0.782	0.738
+BA-SAU	0.901	0.826	0.754	0.709
+RS-DPD +BA-SAU	0.942	0.896	0.812	0.765
HCA-Net (Full)	0.951	0.908	0.833	0.790

TABLE IX

COMPARISON OF PARAMS AND GFLOPs BETWEEN OUR MODEL AND OTHER SOTA MODELS

Methods	Year	Type	Param (M)	GFLOPs (G)
U-Net	2015	CNN	31.044	103.489
U-Net++	2018	CNN	47.176	377.449
PraNet	2020	CNN	30.498	13.150
CaraNet	2021	CNN	44.594	21.749
Polyp-PVT	2021	Transformer	44.983	16.956
HSNet	2022	Transformer	29.257	10.943
PVT-CASCADE	2023	Transformer	35.273	15.404
CAFE-Net	2024	Transformer	35.530	16.119
CIFFormer	2025	Transformer	32.441	13.142
HCA-Net (Ours)	-	Transformer	10.113	9.329

contextual refinement benefits semantic consistency even without additional structural modifications.

When RS-DPD is incorporated into the encoder, a more pronounced performance gain is observed, particularly on the challenging ETIS dataset, suggesting that suppressing deep feature redundancy while preserving structural details significantly enhances cross-dataset generalization. In comparison, BA-SAU also provides stable improvements by enforcing semantic alignment between encoder and decoder features, leading to more accurate boundary localization during decoding.

Notably, combining RS-DPD and BA-SAU yields further improvements beyond either module alone, demonstrating their complementary roles in enhancing both representation discriminability and semantic consistency. The full HCA-Net, which integrates HCA with both RS-DPD and BA-SAU, achieves the best overall performance on both datasets, confirming that the proposed modules work synergistically to balance lightweight design and segmentation accuracy.

G. COMPUTATIONAL EFFICIENCY

Table IX presents the comparison of trainable parameters (Params) and floating-point operations (GFLOPs) across different methods. Traditional CNN-based models, such as U-Net and U-Net++, enhance feature extraction by stacking multiple

convolutional layers, which inevitably results in a large number of parameters and high computational costs. For instance, U-Net++ reaches 47.2 M parameters and 377.4 GFLOPs, making it computationally expensive despite its limited segmentation accuracy. In contrast, PraNet and CaraNet optimize the encoder-decoder structure to reduce FLOPs while maintaining strong segmentation performance. However, CaraNet still involves a relatively large number of parameters due to its ResNet50 backbone (25.6 M parameters). Transformer-based models (e.g., Polyp-PVT, HSNet, and PVT-CASCADE) leverage self-attention mechanisms to capture global context, achieving better parameter efficiency than ResNet50-based networks and offering a more favorable trade-off between accuracy and efficiency. Further comparing Transformer-based approaches, CAFE-Net and CIFFormer maintain parameter counts in the range of 32–36 M, whereas the proposed HCA-Net requires only 10.1 M parameters and 9.3 GFLOPs, which is significantly lower than other counterparts. In addition to requiring only a small number of parameters and low computational cost, HCA-Net runs at approximately 60 FPS on an NVIDIA RTX 4090 GPU with an input resolution of 352×352 , indicating its suitability for real-time applications. This reduction mainly stems from the compact encoder-decoder design of HCA-Net and its lightweight attention mechanism, which combines

TABLE X
COMPARISON WITH REPRESENTATIVE LIGHTWEIGHT POLYP SEGMENTATION MODELS ON THE KVASIR-SEG DATASET

Method	Year	Type	Params (M)	GFLOPs (G)	mIoU	Dice
PolypSeg+	2022	CNN	2.54	7.23	0.706	0.788
DeepNeXt	2024	CNN	1.51	3.04	0.839	0.933
Lite-PolypNet	2025	CNN	8.68	8.63	0.878	0.923
Polyp-LVT	2024	Transformer	25.11	13.21	0.851	0.909
SF-B2	2025	Transformer	27.00	33.00	0.893	0.928
MSFNet	2025	Transformer	9.67	27.90	0.879	0.865
HCA-Net-7	–	Transformer	7.92	6.51	0.847	0.906
HCA-Net-12 (Ours)	–	Transformer	10.11	9.33	0.897	0.936

convolutional projection with adaptive pooling compression to replace redundant multi-head self-attention modules. As a result, HCA-Net effectively reduces both parameters and computation while preserving strong global modeling capability.

To further assess the lightweight characteristics of the proposed method, we compare HCA-Net with representative lightweight polyp segmentation models on the Kvasir-SEG benchmark, including CNN-based approaches (PolypSeg+ [46], DeepNeXt [47], and Lite-PolypNet [48]) and efficient Transformer-based models (Polyp-LVT [35], SF-B2 [49], and MSFNet [50]). As shown in Table X, HCA-Net-12 achieves the best overall segmentation performance among the compared lightweight methods, obtaining 0.897 mIoU and 0.936 Dice with only 10.11 M parameters and 9.33 GFLOPs. Compared with compact CNN-based models, HCA-Net consistently delivers higher segmentation accuracy, indicating that hierarchical contextual modeling provides stronger global–local feature representation than purely convolutional lightweight designs. Compared with lightweight Transformer-based counterparts, HCA-Net achieves competitive or superior accuracy while requiring lower computational cost than heavier attention-based models, demonstrating that its pooled attention formulation effectively captures global context without introducing excessive overhead. Moreover, the smaller variant HCA-Net-7 further reduces computational cost to 6.51 GFLOPs while maintaining reasonable segmentation accuracy, highlighting the flexibility of the proposed framework under strict efficiency constraints. These results confirm that HCA-Net offers a favorable accuracy–efficiency trade-off and is well suited for practical, resource-limited clinical deployment.

V. CONCLUSION

In this paper, we proposed HCA-Net, a lightweight segmentation framework designed to address the common challenges of encoder-decoder architectures, including feature redundancy, semantic misalignment, and insufficient boundary recovery. The framework integrates two complementary modules: the DPD module on the encoder side and the SAU module on the decoder side, both built upon our proposed HCA mechanism. Specifically, HCA-Net introduces a lightweight attention design that integrates local discriminability with global contextual awareness under reduced complexity; DPD alleviates deep feature redundancy while preserving high-fidelity details; and SAU achieves semantic alignment and boundary-sensitive refinement through cross-layer attention. Extensive experiments demonstrate that HCA-Net achieves SOTA accuracy

with significantly improved efficiency, highlighting its clinical applicability.

Although extensive experiments and cross-dataset evaluations are conducted on multiple public polyp segmentation benchmarks, these datasets are widely used in the literature and may share implicit distribution similarities, which could introduce evaluation bias and lead to optimistic generalization estimates. Moreover, the proposed method is specifically designed for 2D colonoscopy image segmentation and does not explicitly model the three-dimensional anatomical structure of the intestinal tract; its extension to 3D volumetric medical imaging data (e.g., CT or MRI) has not yet been explored and would require architectural redesign and additional computational considerations. In addition, the current study does not explicitly address domain shifts caused by differences in imaging devices, acquisition protocols, or hospital-specific settings, which may affect robustness in real-world clinical deployment. Finally, although the design principles of HCA-Net are conceptually general, its effectiveness in non-colonoscopy medical image segmentation tasks remains to be systematically evaluated in future work.

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